

Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat
Department of Computer Science and Engineering
B.Tech II Semester-III
CS231 – Object Oriented Programming Practical Assignment
Lab Assignment - 4

1. Write a C++ program finding maximum and minimum numbers in an array using dynamic memory allocation.
2. Write a C++ program to display and calculate the sum of n numbers by using dynamic memory allocation function.
3. Write a C++ program to read a matrix of size m*n from the keyboard and display the same on the screen using functions.
4. Write a C++ program to swap the values of two numbers by using
 - A. Call by value
 - B. Call by address
 - C. Call by reference variable
5. An electricity board charges the following rates to discourage large consumption of energy.
For the first 100 units -60P per unit
For next 200 units -80P per unit
Beyond 300 units -90P per unit
All users are charged a minimum of Rs. 50.00. If the total amount is more than Rs. 300.00 then an additional surcharge of 15% is added. Write a program to read the name of the user and number of units consumed and print out the charges with names.
6. Write a C++ program to store and display the information of students using structure pointers.
7. Write a C++ program to create three different integer pointers using malloc function and new operator. Then use delete operator and free() function to delete the memory allocated to the above pointers.
For example:

```
int* ptr1 = (int*) malloc (sizeof(int));  
int* ptr2 = new int;  
int* ptr3 = NULL;
```
8. Write a C++ program solving a Tower of Hanoi Problem using a recursive function. **There are 3 disks in the tower 1**
Shift top disk from tower 1 to tower 2; Shift top disk from tower 1 to tower 3;
Shift top disk from tower 2 to tower 3; Shift top disk from tower 1 to tower 2;
Shift top disk from tower 3 to tower 1; Shift top disk from tower 3 to tower 2;
Shift top disk from tower 1 to tower 2.
3 disks in tower 1 are shifted to tower 2.

9. Create a class Student with data members name, roll_no, and a static data member count to keep track of the number of student objects created.

- A. Write a static member function to display the total number of students created.
- B. Demonstrate the program by creating at least 3 objects.

10. Create a class Book with the following:

A. Data members: title, author, and price.

B. A static data member totalBooks to keep track of the total number of books added.

A static member function showTotalBooks() to display the total number of books.

Write a program to demonstrate this by creating at least 4 Book objects, and show the details of each book along with the total number of books created.